

App Summary

Industrial Plant Solutions - Streamlit estimating and operations app

What it is

A Streamlit web app for Industrial Plant Solutions that manages estimating, jobs, time, expenses, users, and assets against a Supabase backend.

The repo also includes proposal export plus AI-assisted material quote import and asset photo autofill features.

Who it's for

Primary persona: internal IPS estimators and admins; viewer users are also supported in auth/session role handling.

What it does

- Builds estimates with materials, labor, equipment, travel, markup, overhead, profit, and tax controls.
- Saves revisions, submits estimates for approval, locks final statuses, and marks awarded work.
- Generates proposal exports as DOCX and PDF, and stores exports and attachments in Supabase Storage.
- Maintains customer and job records, including linked estimates, status, PM/supervisor fields, and awarded amounts.
- Tracks employee time entries with approval status and job/customer context.
- Tracks PO and expense records by category, vendor, job, and status.
- Manages assets, assignments, maintenance, inspections, documents, and AI-assisted photo/quote intake.

How it works

- UI layer: ``app/main.py`` registers Streamlit pages and ``app/ui.py`` renders custom sidebar navigation.
- Auth/session: ``app/auth.py`` signs users in through Supabase Auth, loads ``profiles``, and gates admin-only pages by role.
- Data access: ``app/db.py`` wraps Supabase table CRUD, admin writes, signed URLs, storage upload, and quote number generation.
- Feature pages: estimating, jobs, time, expenses, materials, labor, equipment, users, and asset pages read/write Supabase tables directly.
- Services: ``app/services/material_quote_import.py`` and ``app/services/asset_photo_autofill.py`` call OpenAI for structured extraction from uploaded files/photos.
- Schema evidence: ``sql/001_asset_module_tables.sql`` defines asset tables; other non-asset table DDL was not found in repo.

How to run

- From the project root, create ``.env`` values for ``SUPABASE_URL`` plus either `publishable/anon` and `secret/service-role` keys; ``OPENAI_API_KEY`` is also needed for AI features.
- Install Python dependencies: Not found in repo (``requirements.txt`` / top-level ``pyproject.toml`` not found).
- Start the app with ``streamlit run app/main.py``.

Source basis: repo files only. Visual PNG rendering was not available in this environment because ``pdftoppm`` was not installed.